

Hari Prasath J

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Summary

Final year B.Tech student in Artificial Intelligence & Data Science with strong foundations in **Data Structures and Algorithms**, demonstrated by solving 250+ problems across arrays, trees, graphs, hashing, and dynamic programming. Proficient in **Java and Python**, with hands-on experience building scalable backend systems using Spring Boot and SQL. Seeking a Software Engineering Internship to contribute strong problem-solving and backend development skills.

EDUCATION

B.Tech in Artificial Intelligence and Data Science

Sri Krishna College of Engineering and Technology • Coimbatore , Tamil Nadu • GPA: 8.28 • 09/2027

PROJECTS

RetiScan

- **Diabetic Retinopathy Classification Engine:** Built a deep learning pipeline utilizing an optimized CNN architecture and a custom **Focal Loss** function to classify 5 progressive severity stages from fundus imagery.
- **Automated CV Pre-Screening Filter:** Engineered a real-time computer vision quality control pipeline using adaptive thresholding and contour-fitting (`minEnclosingCircle`) to validate image exposure and isolate structural regions-of-interest.
- **Explainable AI (XAI) Quantification:** Developed an XAI localization module integrating **Grad-CAM maps** and a mathematical **Attention Intensity Index** to visually isolate and quantify active lesion quadrants.
- **Longitudinal Patient Dashboard:** Deployed a **Streamlit micro-service** with a JSON database to securely graph patient diagnostic trends across multiple clinic visits.

Sentiment analyser app

- Built a supervised NLP classification system to analyze product-related tweets into positive and negative sentiments.
- Applied tokenization and vectorization techniques using Python libraries.

Dog breed classification system

- Developed a deep learning model using **Keras (TensorFlow backend)** to classify dog breeds across 120 categories, achieving 97% validation accuracy.
- Implemented image preprocessing and model optimization techniques for multi-class classification and hosted the model using Streamlit to enable interactive image uploads and live inference.

Legal workflow guidance

- Built a full-stack Indian legal guidance platform using Flask, FAISS semantic search, and LLaMA 3.3-70B (via Groq), dynamic intake interviews, and AI-generated complaint drafts with PDF export.
- Designed a RAG-based legal case classifier with a SentenceTransformer embedding index over a custom knowledge base, enabling real-time case matching, evidence strength analysis, and sensitive case detection with trauma-informed intake flows.

SKILLS

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|----------------------------------|-------------------------------------|----------------------|-------------|
| • Java | • Object-Oriented Programming (OOP) | • AWS Cloud Services | • Streamlit |
| • Data Structures and Algorithms | • SQL | • Python | • CNN |
| | | • DBMS | |

CERTIFICATIONS

AWS cloud practioner

Enterprise networking and security

Introduction to CyberSecurity

Cloud Computing

Python essentials